NUNO ZHAN

BSc Physics (Year 2)—University of Edinburgh
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PROFILE

Second Year BSc Physics student at The University of Edinburgh with experience in experimental laboratory work, computational simulations and mathematical modelling. Strong interest in applied physics in aerodynamics and fluid dynamics. Multilingual in English, Spanish and Mandarin Chinese.

EDUCATION

BSc Physics (Hons) September 2024 - present

Year 2 (entering third year after summer) and currently averaging 74%.

Relevant coursework:

1. **Experimental Physics 2** (72%, A3): Experimental laboratory course with determination of the acceleration due to gravity using compound and Kater's pendulum, Michelson Interferometer and the management of data analysis and error propagation
2. **Dynamics and Vector Calculus**: applied vector calculus to classical mechanics problems including irrotational fields, Stokes' Theorem, and multidimensional integration.
3. **Computer Simulation** (89%, A2): Mandelbrot Set, Simple Cellular automaton, Monte Carlo simulation, final project simulating an animated chaotic N-body system (code available in Github)

Spanish Baccalaureate (Bachillerato): Honors Excellence Award September 2022 - July 2024

RESEARCH & TECHNICAL PROJECTS

Aerofoil Flow Simulation (Summer 2026, ongoing)

- Implementing potential flow methods including source/sink, doublet, and vortex elements to model 2D aerodynamic flows
- Developing a vortex-source panel method to simulate pressure distribution.
- Applying the Method of Images and vortex lift theory to extend simulations to ground effect and lifting surfaces.

Solar System Simulation

- Built and animated chaotic N-body gravitational simulation using Python and related scientific packages
- Implemented the Beeman method as the main numerical integration method to simulate the whole planetary motion over long timescales.
- Compared with Euler and Euler-Cromer algorithms to check the accuracy of the higher order method
- Detected planetary alignments by computing mean angular positions of each planet over century-long timescales using SciPy.

Stochastic Epidemic Modelling - Disease Spread Simulation

- Developed a Python-based simulation to model SIRS dynamics, incorporating re-infection variables and second wave detection logic
- Combined systems of differential equations with a stochastic Monte Carlo model to better reflect realistic dynamics through random event timings.
- Analysed of the statistical probability of epidemic extinction by visualising end time distributions from frequency histograms.

Michelson Interferometer - Sodium Doublet Wavelength Splitting

- Calibrated the micrometer gearing ratio using a Mercury lamp, then determined mean wavelength and doublet splitting by counting fringe shifts and blur transitions
- Performed full uncertainty analysis including systematic and random error sources.
- Achieved results deviating by 0.22% from the accepted literature range for the mean wavelength, and 0.30% for the doublet splitting.

INTERNSHIP/EXPERIENCE

Sputnik Program - *Entrepreneurship and Future Tech Scholar* Sept 2022
Selected for a corporate-sponsored leadership bootcamp; focused on start-up scalability and the practical application of blockchain in business for data storage and company future expansion.

ARCyTAN Program - *Physics and Dynamics Awardee (Bronze Medalist)* Dec 2023
Ranked as a Bronze Medalist in regional physics; qualified for Spanish Physics Olympiads and completed technical seminars at the School of Engineering (ETSi)

European Youth Parliament (EYP) - *Delegate and Presenter* Mar 2023
Selected to research Political Justice; drafted and defended a resolution for the JURI II committee as a formal presenter during the assembly

Sales Associate - *Customer Relations* Summer Work 2022–2024
Worked busy shifts across two stores explaining technical product details under high-pressure, high-volume conditions

TECHNICAL SKILLS

- **Microsoft Suite:** Excel, Word, PowerPoint, Outlook
- **Programming:** Python (Object Oriented Programming, numpy, scipy, sympy, matplotlib, vectorisation)
- **Simulation and Modelling:** Monte Carlo, chaotic N-body, ODE solvers and numerical integration methods.
- **Research:** LaTeX report writing, mathematical modelling, scientific instrumentation, error and statistical analysis

ADDITIONAL INFORMATION

- Enjoy communicating complex concepts to both technical and non-specialist audiences.
- Languages: English (fluent), Spanish (native), Chinese (native), French (Upper-Intermediate)
- Player in the University of Edinburgh Men's Volleyball Team, currently playing in BUCS League and Scottish Plate